Open problems:
Entropy is not continuous in standard probability spaces

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **The appropriate topology on P_μ.** The brief raises "the question as to what is the appropriate one for probability spaces" and shows only that a topology making mixing and S continuous must refine total variation. Whether such a topology exists with good properties (e.g. Hausdorff), and which one is the coarsest, is left open. (Brief, Sec. 1; Theorem 46; Sec. 5.)

2. **When is entropy L^1-continuous?** Theorem 21 needs disjoint sets of arbitrarily large finite measure; the conclusion extends the failure to "typical infinite-dimensional spaces". Measure spaces outside the hypothesis, such as Lebesgue measure on $[0, 1]$, are not treated, and no characterization of the spaces on which S is L^1-continuous is given. (Brief, Theorem 21; Sec. 5.)

From the formalization

1. **The Jensen–Shannon topology as a topology.** Prop. 10 is formalized through convergence along filters; the "topology generated by the Jensen–Shannon divergence" is not constructed, and since JSD is not a metric it is not shown that its balls generate a topology equal to the $L^1(\mu)$ topology. (Log, "Differences from the brief".)

2. **Total variation identity.** Eq. (2), $\|p_\rho - p_\sigma\|_{\mathrm{TV}} = \frac{1}{2}\|\rho - \sigma\|_{L^1(\mu)}$, and with it the passage from L^1 convergence to total-variation convergence of measures, is not formalized. (Log, "Differences from the brief".)

3. **P_μ closed under all mixtures.** Only the equal mixture $(\rho + \sigma)/2 \in P_\mu$ is proved. That $\lambda\rho + (1 - \lambda)\sigma \in P_\mu$ for all $\lambda \in [0, 1]$, which the brief's mixing hypothesis in Theorem 46 presupposes, is not formalized. (Log, "Problems found in the brief".)